



The University of Texas at Austin

Design and Construction Standards

Technical Specification

SECTION 23 37 00

Air Inlets and Outlets

All renovation and new construction activities on The University of Texas at Austin campuses are required to meet the UT Austin Design and Construction Standards (DCS) from the planning and design stages through actual construction and facilities maintenance. These standards reflect the planning, design, construction, maintenance, and other facilities asset expertise of University personnel. Any exceptions shall be submitted by the Project Manager as a Standards Exception Request and incorporated after approval is received from the respective Division Champion. The Design and Construction Standards are managed by Campus Operations, and updated with the advice of multiple campus groups at The University of Texas at Austin. This document is the property of UT Austin, and use of this document, in part or in whole, for any purpose other than for a UT Austin project may not be done without written permission of the University.

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The issuance and revision history of this Section is tabulated below. Please destroy any previous copy in your possession.

Rev Date	Pages	Remarks	Documents Referenced
3/26/2021	All	New Technical Specification created.	
6/30/2022	All	New Technical Specification published.	
9/1/2025	3	Minor Edit – 2.1.A – add Titus as an acceptable manufacturer.	

SECTION 23 37 00 – AIR INLETS AND OUTLETS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Perform all Work required to provide and install diffusers, diffuser boots, registers/grilles, louvers, louver penthouses, roof hoods, and goosenecks indicated by the Contract Documents with supplementary items necessary for proper installation.

1.3 REFERENCE STANDARDS

- A. The latest published edition of a reference shall be applicable to this Project unless identified by a specific edition date.
- B. All reference amendments adopted prior to the effective date of this Contract shall be applicable to this Project.
- C. All materials, installation and workmanship shall comply with the applicable requirements and standards addressed within the following references:
 - 1. AMCA 500 – Test Method for Louvers, Dampers and Shutters.
 - 2. ANSI/NFPA 90A – Installation of Air Conditioning and Ventilating Systems.
 - 3. AHRI 890 – Rating of Air Diffusers and Air Diffuser Assemblies.
 - 4. ASHRAE 70 – Method of Testing for Rating the Air Flow Performance of Outlets and Inlets.
 - 5. SMACNA 1035 – HVAC Duct Construction Standards - Metal and Flexible.

1.4 QUALITY ASSURANCE

- A. Test and rate performance of air outlets and inlets in accordance with ASHRAE 70.
- B. Test and rate performance of louvers in accordance with AMCA 500.

1.5 SUBMITTALS

- A. Product Data:
 - 1. Submit product data and Shop Drawings, indicating type, size, location, application, material, finish, and type of mounting. Submit performance data including throw and drop, static pressure drop and noise ratings.
 - 2. Review requirements of outlets and inlets as to size, finish, and type of mounting prior to submitting product data.
- B. Operation and Maintenance Data:
 - 1. Submit manufacturer's installation instructions under provisions of Division 01.

- C. Samples: At the request of the Owner and/or A/E team, submit each exposed product for each color and texture specified.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Products meeting all requirements of this specification Section of the following manufacturers are acceptable:
 - 1. Krueger
 - 2. Price
 - 3. Metalaire
 - 4. Titus
 - 5. Approved equal

2.2 GENERAL:

- A. Grilles, registers and ceiling outlets shall be as scheduled on the Drawings and shall be provided with sponge rubber or soft felt gaskets. If a manufacturer other than the one scheduled is used, the sizes shown on the Drawings shall be checked for performance, noise level, face velocity, throw, pressure drop, etc., before the submittal is made. Selections shall meet the manufacturer's own published data for the above performance criteria.
- B. For non-laboratory or laminar flow diffusers: The throw shall be such that the velocity at the end of the throw in the five-foot occupancy zone will be not more than 50 FPM nor less than 25 FPM. Noise levels shall not exceed those published in the ASHRAE Guide for the type of space being served (NC level).
- C. For laboratory or laminar flow diffusers: The discharge face velocity shall not be not more than 40 FPM. Noise levels shall not exceed those published in the ASHRAE Guide for the type of space being served (NC level).
- D. Locations of outlets on Drawings are approximate and shall be coordinated with other trades to make symmetrical patterns, and shall be governed by the established pattern of the lighting fixtures or architectural reflected ceiling plan. Where called for on the schedules, the grilles, registers and ceiling outlets shall be provided with deflecting devices and manual damper. These shall be the standard product of the manufacturer, subject to review by the Architect, and equal to brand scheduled.

2.3 ROUND CONE CEILING DIFFUSERS

- A. Provide aluminum round, adjustable discharge pattern, stamped or spun, multicore type diffuser with neck sizes, dimensions, and mounting types as shown on plans and schedules.
- B. Air pattern: 360° horizontal to vertical adjustable discharge pattern. Pattern shall be adjustable by means of moving cores up or down.
- C. Construction: Four all aluminum, seamless spun cones with repositioning screws to adjust inner cones to change air pattern.
- D. Extend diffuser collar not more than one inch above ceiling face and connect to the duct with duct ring. In plaster ceilings, provide plaster ring and ceiling plaque.

- E. Provide with white powder coat finish unless otherwise indicated.

2.4 SQUARE PLAQUE CEILING SUPPLY DIFFUSERS

- A. Provide aluminum square architectural plaque ceiling diffusers with neck sizes, dimensions, and mounting types as shown on plans and schedules.
- B. Air Pattern: 360° horizontal air flow pattern with strong Coanda effect.
- C. Backpan: Diffuser shall have seamless formed aluminum back cone with square or round inlet collar for duct connections dimensioned as scheduled.
- D. Frame: Provide frame type appropriate for ceiling type. For lay in ceilings, diffuser shall fit in inverted T-bar frame. For plaster ceilings, diffuser shall fit in plaster frame and ceiling frame.
- E. Face Panel: The face panel of the diffuser shall be smooth, flat, and free of visible fasteners. The face panel shall project no more than 1/4" below the outside border of the diffuser backpan. The face panel shall be removable from the diffuser face.
- F. Provide with white powder coat finish unless otherwise indicated.

2.5 PERFORATED FACE CEILING SUPPLY DIFFUSERS

- A. Provide diffuser with fully adjustable pattern, and removable aluminum perforated face.
- B. Air Pattern: Horizontal air flow pattern with strong Coanda effect. Pattern controllers shall be mounted on the back of the perforated face and must be field adjustable to allow discharged air to enter the room in either vertical or one, two, three, or four-way jets.
- C. Backpan: Steel backpan with inlet collars dimensioned as scheduled.
- D. Frame: Provide steel frame with type appropriate for ceiling type. For lay in ceilings, diffuser shall fit in inverted T-bar frame. For plaster ceilings, diffuser shall fit in plaster frame and ceiling frame.
- E. Face Panel: Provide perforated face with minimum 51% free area. Face shall be removable from the backpan.
- F. Provide with white powder coat finish unless otherwise indicated.

2.6 PERFORATED FACE CEILING RETURN/EXHAUST GRILLES

- A. Provide grille with fully adjustable pattern, and removable aluminum perforated face.
- B. Backpan: Steel backpan with inlet collars dimensioned as scheduled.
- C. Frame: Provide steel frame with type appropriate for ceiling type. For lay in ceilings, diffuser shall fit in inverted T-bar frame. For plaster ceilings, diffuser shall fit in plaster frame and ceiling frame.
- D. Face Panel: Provide perforated face with minimum 51% free area. Face shall be removable from the backpan.
- E. Provide with white powder coat finish unless otherwise indicated.

2.7 RADIAL LABORATORY SUPPLY DIFFUSER

- A. Provide radial flow diffuser with stainless steel backpan and hinged stainless steel perforated face. Face shall be flush faced or shall descend 5 3/4" below the ceiling as scheduled.

- B. Air Pattern: One or two-way radial discharge with low discharge velocity to minimize mixing/entrainment of room air.
- C. Mounting Frames: For lay in ceilings, diffuser shall fit in inverted T-bar frame. For plaster ceilings, diffuser shall fit in plaster frame and ceiling frame.
- D. Plenum: Stainless steel plenum. Provide integral hanger tabs for securing the unit to overhead structure. Provide with foam gasketing at face/backpan interface. Provide inlet collars dimensioned as scheduled.
- E. Face Panel: Provide perforated face with engineered deflection vanes. Face shall be provided with piano hinges and tool-less quick release latches to allow access for cleaning.
- F. Provide with #4 polish finish unless otherwise indicated.

2.8 LAMINAR FLOW SUPPLY DIFFUSER

- A. Provide non-aspirating, unidirectional type laminar flow diffuser.
- B. Air Pattern: Unidirectional, non-aspirating airflow with low discharge velocity.
- C. Mounting Frames: For lay in ceilings, diffuser shall fit in inverted T-bar frame. For plaster ceilings, diffuser shall fit in plaster frame and ceiling frame.
- D. Plenum: Continuously welded stainless-steel plenum with internal pressure equalization baffle to create uniform face velocity across the grille. Provide integral hanger tabs for securing the unit to overhead structure. Provide with foam gasketing at face/backpan interface. Provide inlet collars dimensioned as scheduled.
- E. Face Panel: Provide perforated face. Face shall be provided with tool-less quick release latches to allow access for cleaning, anti-slip, snap-in retainers, and stainless-steel retainer cables.
- F. Provide with #4 polish finish unless otherwise indicated.

2.9 SUPPLY SIDEWALL REGISTERS/GRILLES

- A. Provide aluminum sidewall grilles with streamlined and individually adjustable curved blades to discharge air along face of grille, with one-way or two-way deflection as scheduled.
- B. Blades: Provide one or two sets of fully adjustable blades (as scheduled) with $\frac{3}{4}$ " on center spacing.
- C. Provide with white powder coat finish unless otherwise indicated.
- D. Provide integral galvanized steel gang-operated opposed blade damper with removable key operator, operable from face where indicated.

2.10 EXHAUST AND RETURN SIDEWALL REGISTERS/GRILLES

- A. Provide aluminum sidewall louver type grille with streamlined fixed curved blades.
- B. Blades: Provide one set of fixed blades with $\frac{3}{4}$ " on center spacing with 45° deflection.
- C. Provide with white powder coat finish unless otherwise indicated.
- D. Provide integral galvanized steel gang-operated opposed blade damper with removable key operator, operable from face where indicated.

2.11 CEILING SLOT DIFFUSERS

- A. Provide continuous slot diffusers with direction vanes and slot widths as scheduled. The complete slot diffuser system shall include engineered and factory-built plenums, hoods, blankoffs, and associated components. All components shall be made by a single manufacturer.
- B. The slot diffusers shall integrate into the ceiling system. Where curved linear slot diffusers are indicated, they shall be stretch formed to the exact curve required.
- C. The slot diffusers shall have a single slot unless scheduled otherwise and shall be capable of being used for supply, return, exhaust air, or any combination thereof.
- D. Linear diffusers supported by screws in flanges or from air plenums are unacceptable. For lay-in ceilings, provide wire hanger support clips integral with the linear slot diffusers allowing the slot diffusers to be supported from the building structure with ceiling wire. For hard ceilings, provide clips that are integral with the slot diffusers allowing the diffusers to be secured directly to the ceiling framing without hanger supports. Provide spline clips to secure joints and ceiling tees to the diffusers.
- E. The slot diffuser systems shall include an engineered plenum and pattern controllers designed to control the throw from the diffuser.
- F. Slot diffusers shall be fabricated of aluminum extrusions with factory baked enamel finish.

PART 3 - EXECUTION**3.1 INSTALLATION**

- A. Install items in accordance with manufacturers' instructions.
- B. Check location of outlets and inlets and make necessary adjustments in position to conform with architectural features, symmetry, and lighting arrangement.
- C. Install diffusers to ductwork with air tight connection.
- D. Provide balancing dampers on duct take-off to diffusers, and grilles and registers, regardless of whether dampers are specified as part of the diffuser, or grille and register assembly.
- E. Do not install ceilings adjacent to fixtures until installation of fixtures, air supply assemblies, return-air blank-off strips and flexible duct have been properly approved. Remove and reinstall any part of the installation found incorrect.
- F. Diffusers. Louvered diffuser outlets mount tight against the ceiling. Fasten outlets securely to ductwork with sheet metal screws. For perforated diffusers, attach the frame assembly by a concealed hinge assembly to an outer frame compatible with the type of ceiling on which the diffuser is installed.
- G. Radial Diffusers. Install per manufacturer's recommendations and as indicated in details. All radial diffusers to be independently hung from structure.
- H. Paint ductwork visible behind air outlets and inlets matte black.
- I. Blank Offs. Install diffusers per drawings for directional pattern.

3.2 ADJUSTING

- A. After installation, adjust diffusers, registers, and grilles to air patterns indicated, or as directed, before starting air balancing.

END OF SECTION 23 37 00